LAB 1

Requirements Engineering & UML Use-Case Modelling

Vaccination Cohort & Dose Scheduling System
Problem Statement #17 — Healthcare & Telemedicine

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1. Project Overview

The Vaccination Cohort & Dose Scheduling System is a public-health administration platform designed to organize demographic vaccination rollouts, enforce minimum intervals between doses, schedule vaccination-center appointments, maintain vaccination records, and issue verifiable QR digital certificates.

Primary actors

• Citizen Registrant — registers, checks eligibility, books and manages appointments, and accesses certificates.
• Vaccination Officer — manages center capacity, views appointments, records administered doses, and supports certificate-related operations.

2. Functional Requirements

Exactly five functional requirements are specified below.

ID	Requirement	Description	Priority	Acceptance Criteria	Rationale
FR-001	Citizen Registration & Cohort Assignment	The system shall allow a citizen to register by providing required identity and demographic details, validate the submitted information, and assign the citizen to the appropriate vaccination cohort.	High	Given valid mandatory details, the system creates one citizen profile and assigns the correct cohort; invalid or duplicate registration is rejected with an explanation.	Establishes the citizen record required for all subsequent vaccination activities.
FR-002	Dose Eligibility & Interval Validation	The system shall display a citizen's dose status and determine eligibility for the next dose. For Dose 2, the system shall enforce the configured minimum interval of 28 days after Dose 1 before allowing booking.	High	If fewer than 28 days have elapsed after Dose 1, Dose 2 booking is blocked; once the interval is met and other rules pass, the citizen can proceed.	Prevents premature or invalid vaccination and follows the scenario's explicit interval rule.
FR-003	Vaccination Slot Booking & Appointment Management	The system shall allow eligible citizens to view available vaccination-center slots and book, view, or reschedule an appointment, subject to slot capacity.	High	A booking is confirmed only when the citizen is eligible and the selected slot has capacity; the system updates slot availability and displays the appointment details.	Supports orderly scheduling and avoids overbooking at vaccination centers.
FR-004	Vaccination Administration & Record Management	The system shall allow an authorized vaccination officer to view scheduled appointments, record an administered dose, and update the citizen's vaccination history.	High	Only an authenticated officer can record a dose; a successful record stores the dose number, date, center, and officer reference and updates the citizen's status.	Maintains an accurate and auditable vaccination history.
FR-005	QR Certificate Generation & Verification	The system shall generate a digitally signed QR vaccination certificate after a dose is recorded and provide a verification function that validates the certificate's authenticity and status.	High	A certificate is generated only for a recorded vaccination; verification accepts a valid signature and rejects altered, invalid, or revoked certificates.	Provides citizens with portable proof and enables trusted verification by authorized parties.

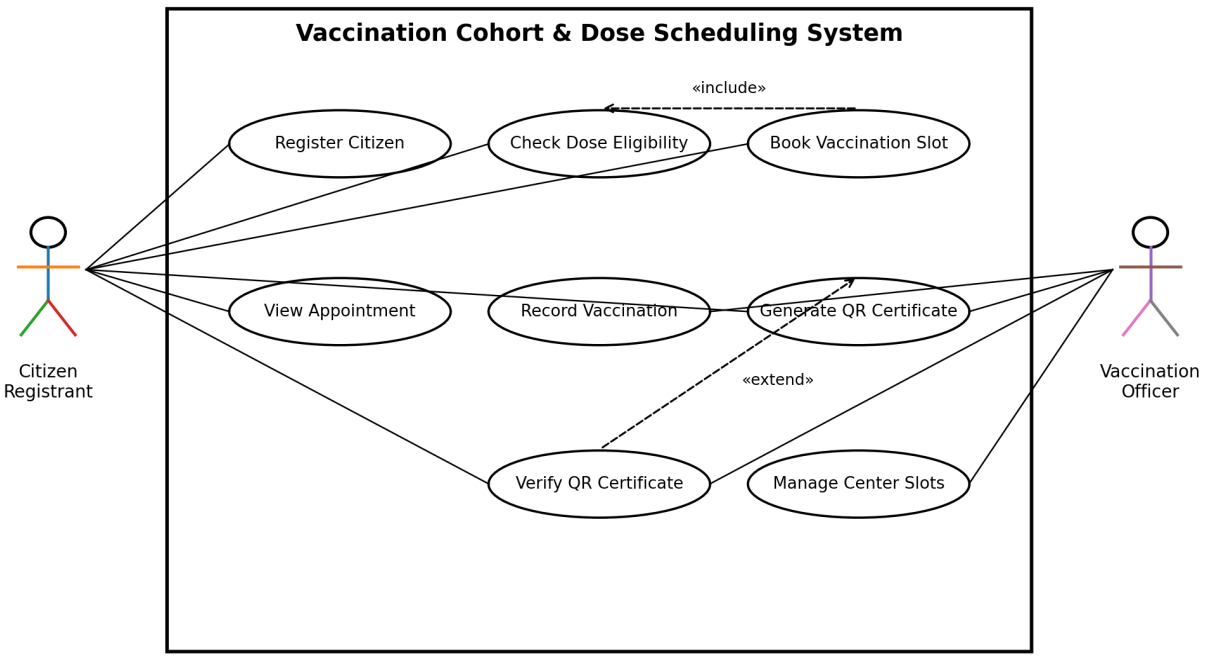
3. Non-Functional Requirements

Exactly two non-functional requirements are specified below.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The certificate verification endpoint shall validate digitally signed QR certificates in under 150 milliseconds under the defined normal test load, and shall use cryptographic signature verification.	High	A benchmark test confirms verification latency below 150 ms at the agreed test load; modified or incorrectly signed certificate payloads are rejected.	The supplied problem statement explicitly identifies sub-150 ms verification and digitally signed certificates.
NFR-002	Privacy & Access Control	The system shall protect personal and vaccination data using authenticated access, role-based authorization, and encryption in transit and at rest.	High	Unauthenticated requests cannot access protected records; citizens can access only their own records, while vaccination officers can perform only authorized operational actions; data is encrypted in transit and at rest.	The system handles sensitive health and identity information and must prevent unauthorized disclosure or modification.

4. UML Use-Case Model

The use-case model identifies the two primary actors and the system functions they interact with. «Include» represents mandatory reused behaviour; «extend» represents optional or conditional behaviour.



Dashed dependency labels identify mandatory included behaviour and optional extension.

Relationship notes:

• Book Vaccination Slot «includes» Check Dose Eligibility — eligibility must be validated before a booking is confirmed.
• Verify QR Certificate is modelled as an optional follow-on action associated with certificate issuance and use.

5. Use-Case Flow Specification

Use Case: Book Vaccination Appointment

Primary Actor: Citizen Registrant
Goal: Reserve an available vaccination-center slot for an eligible dose.
Preconditions: The citizen has a registered account and is authenticated. The vaccination center has published its slots.
Postconditions: A confirmed appointment is stored, the selected slot capacity is reduced, and the citizen can view the appointment details.

Main Success Scenario

1. Citizen opens the appointment booking function.
2. System authenticates the citizen and retrieves the citizen profile and vaccination history.
3. Citizen selects the required dose and preferred vaccination center.
4. System checks dose eligibility, including the minimum 28-day interval for Dose 2 where applicable.
5. System retrieves available slots for the selected center and date.
6. Citizen selects an available slot and confirms the booking.
7. System reserves the slot, updates remaining capacity, and creates the appointment record.
8. System displays the booking confirmation and appointment details.

Alternate Flows

• If authentication fails, the system asks the citizen to sign in again and does not expose personal records.
• If the citizen is not eligible for the selected dose, the system explains the reason (for example, the 28-day interval has not elapsed) and prevents booking.
• If the selected slot becomes unavailable before confirmation, the system informs the citizen and displays other available slots; no duplicate booking is created.

6. Traceability to Lab 2

These requirements provide the basis for the Agile backlog prepared in Lab 2. The eight Jira stories cover citizen registration and cohort eligibility (FR-001), dose eligibility and interval-gated booking (FR-002), slot viewing and center capacity (FR-003), vaccination recording (FR-004), and QR certificate generation and verification (FR-005). The certificate verification performance target in NFR-001 is retained as a system quality constraint.

Lab 2 used four Epics and eight User Stories (SCRUM-11 to SCRUM-18), with 37 total story points. The requirements above are consolidated to match those backlog themes.

7. Submission Checklist

- ■ Exactly 5 Functional Requirements (FR-001 to FR-005)
- ■ Exactly 2 Non-Functional Requirements (NFR-001 and NFR-002)
- ■ UML Use-Case Diagram with actors and «include» / «extend» relationships
- ■ One core Use-Case Flow with preconditions, postconditions, main success scenario, and alternate flows